

Phase 1 — Autoencoder Losses

Reconstruction $\mathcal{L}_{\text{recon}}$

$$\text{MSE}(\text{dec}(f_\psi(z_t)), x_t)$$

every t , incl. z_t resampled via VAE reparam



Next-frame prediction $\mathcal{L}_{\text{recon_next}}$

$$\text{MSE}(\text{nfd}(h_t, a_t), x_{t+1})$$

forces h_t to be encode angular velocity as well as position



KL to unit prior $\mathcal{L}_{\text{kl}}$

$$[-\frac{1}{2}(1 + \log \sigma^2 - \mu^2 - \sigma^2)]_{\geq \text{free_bits}}$$

summed over latent dims, mean over batch/time



Temporal metric reg. $\mathcal{L}_{\text{temporal}}$ (*opt.*)

$$\text{ReLU}(c|t_1 - t_2| - \|h_{t_1} - h_{t_2}\|)$$

random pairs; one-sided, only penalises being *too close*



Sparsity $\mathcal{L}_{\text{sparsity}}$ (*opt.*)

$$\|\mu\|_1 \text{ (L1 on latent mean)}$$

pushes unused h -dims to exactly 0



Phase 1 total

$$\mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{recon_next}} + w_{\text{kl}}\mathcal{L}_{\text{kl}}$$

$$(+ w_{\text{temp}}\mathcal{L}_{\text{temporal}} + w_{\text{sp}}\mathcal{L}_{\text{sparsity}})$$

Phase 2 — Dynamics Losses

Log-det regulariser $\mathcal{L}_{\text{logdet}}$

$$\mathbb{E}[(\log |\det J_\Phi|)^2] \text{ over window}$$

keeps Φ volume-preserving-ish / well-conditioned



Teacher-forced $\mathcal{L}_{\text{tf}}$

$$\text{MSE}(\Phi^{-1}(\text{RK4}(\Phi(h_t), u_t)), h_{t+1})$$

one step, every t in window — batched, no sequential graph



Closed-loop rollout $\mathcal{L}_{\text{cl}}$

$$\sum_{k=1}^T \frac{\gamma^{k-1}}{\sum_j \gamma^{j-1}} \text{MSE}(\Phi^{-1}(q_k, p_k), h_{\text{ctx}+k})$$

k Hamiltonian steps from one seed, no re-encoding;
 γ -discounted so near-term error dominates



Hamiltonian L1 / structural reg. (*opt.*)

$$\|\theta_H\|_1, \quad \|R_{pp}\|_F^2$$

sparsify H_γ 's params; damp the learned R block



Phase 2 total

$$\mathcal{L}_{\text{logdet}} + w_{\text{tf}}\mathcal{L}_{\text{tf}} + w_{\text{cl}}\mathcal{L}_{\text{cl}}$$

$$(+ w_{\ell 1}\|\theta_H\|_1 + w_{\text{struct}}\|R_{pp}\|_F^2)$$